COHEN–MACAULAY GROUPOID SLICING LOWERS ORBIT-FIBRE DIMENSION BY ONE

THE STACKS PROJECT AUTHORS

Editorial note. In the displayed intermediate dimension quotient the author writes f; in the target local ring this means t(f), as explicitly used in the subsequent construction. The missing closing parenthesis and occasional i/j minimal-prime index slip are typographical and do not change the stated construction.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Groupoid restriction and stabilizer base change, dimension of locally finite-type groupoids over a field, prime avoidance, dimension formula for Cohen–Macaulay local rings, and the standard flatness criterion for a fibrewise nonzerodivisor.

Source reference. Original definition 02J1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3320–3320. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02J2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3321–3321. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02FX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3208–3208. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6050–6050. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04MQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6051–6051. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04LU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6020–6020. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:more-groupoids.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04ML: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6047–6047. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00NA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 852–852. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00DS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 508–508. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00KW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 766–766. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02JN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3342–3342. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 046Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5490–5490. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 04MN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6049–6049. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let S be a scheme. Let (U, R, s, t, c, e, i) be a groupoid scheme over S. Let $G \to U$ be the stabilizer group scheme. Assume s and t are Cohen-Macaulay and locally of finite presentation. Let $u \in U$ be a finite type point of the scheme U, see Morphisms, Definition 02J1. With notation as in Situation 0.2, set*

$$d_1 = \dim(G_u), \quad d_2 = \dim_{e(u)}(F_u).$$

If $d_2 > d_1$, then there exist an affine scheme U' and a morphism $g : U' \to U$ such that (with notation as in Situation 0.2)

1. *g is an immersion*
2. *$u \in U'$,*
3. *g is locally of finite presentation,*
4. *the morphism $h : U' \times_{g,U,t} R \longrightarrow U$ is Cohen-Macaulay at $(u, e(u))$, and*
5. *we have $\dim_{e'(u)}(F'_u) = d_2 - 1$.*

Situation 0.2. Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Let $g : U' \to U$ be a morphism of schemes. Let $u \in U$ be a point, and let $u' \in U'$ be a point such that $g(u') = u$. Given these data, denote (U', R', s', t', c') the restriction of (U, R, s, t, c) via the morphism g. Denote $G \to U$ the stabilizer group scheme of R, which is a locally closed subscheme of R. Denote h the composition

$$h = s \circ \mathrm{pr}_1 : U' \times_{g,U,t} R \longrightarrow U.$$

Denote $F_u = s^{-1}(u)$ (scheme theoretic fibre), and G_u the scheme theoretic fibre of G over u. Similarly for R' we denote $F'_{u'} = (s')^{-1}(u')$. Because $g(u') = u$ we have

$$F'_{u'} = h^{-1}(u) \times_{\mathrm{Spec}(\kappa(u))} \mathrm{Spec}(\kappa(u')).$$

The point $e(u) \in R$ may be viewed as a point on G_u and F_u also, and $e'(u')$ is a point of R' (resp. $G'_{u'}$, resp. $F'_{u'}$) which maps to $e(u)$ in R (resp. G_u, resp. F_u).

Proof. Let $\mathrm{Spec}(A) \subset U$ be an affine neighbourhood of u such that u corresponds to a closed point of U, see Morphisms, Lemma 02J2. Let $\mathrm{Spec}(B) \subset R$ be an affine neighbourhood of $e(u)$ which maps via j into the open $\mathrm{Spec}(A) \times_S \mathrm{Spec}(A) \subset U \times_S U$. Let $\mathfrak{m} \subset A$ be the maximal ideal corresponding to u. Let $\mathfrak{q} \subset B$ be the prime ideal corresponding to $e(u)$. Pictures:

$$\begin{array}{ccc} B & \xleftarrow[s]{} & A \\ {\scriptstyle t}\uparrow & & \\ A & & \end{array} \quad \text{and} \quad \begin{array}{ccc} B_{\mathfrak{q}} & \xleftarrow[s]{} & A_{\mathfrak{m}} \\ {\scriptstyle t}\uparrow & & \\ A_{\mathfrak{m}} & & \end{array}$$

Note that the two induced maps $s, t : \kappa(\mathfrak{m}) \to \kappa(\mathfrak{q})$ are equal and isomorphisms as $s \circ e = t \circ e = \mathrm{id}_U$. In particular we see that $\mathfrak{q}$ is a maximal ideal as well. The ring maps $s, t : A \to B$ are of finite presentation and flat. By assumption the ring

$$\mathcal{O}_{F_u, e(u)} = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$$

is Cohen-Macaulay of dimension d_2. The equality of dimension holds by Morphisms, Lemma 02FX.

Let R'' be the restriction of R to $u = \mathrm{Spec}(\kappa(u))$ via the morphism $\mathrm{Spec}(\kappa(u)) \to U$. As $u \to U$ is locally of finite type, we see that $(\mathrm{Spec}(\kappa(u)), R'', s'', t'', c'')$ is a groupoid scheme with s'', t'' locally of finite type, see Lemma 04MP. By Lemma 04MQ this implies that $\dim(G'') = \dim(R'')$. We also have $\dim(R'') = \dim_{e''}(R'') = \dim(\mathcal{O}_{R'', e''})$, see Lemma 04LU. By Groupoids, Lemma 04ML we have $G'' = G_u$. Hence we conclude that $\dim(\mathcal{O}_{R'', e''}) = d_1$.

As a scheme R'' is

$$R'' = R \times_{(U \times_S U)} \Big(\mathrm{Spec}(\kappa(\mathfrak{m})) \times_S \mathrm{Spec}(\kappa(\mathfrak{m})) \Big)$$

Hence an affine open neighbourhood of e'' is the spectrum of the ring

$$B \otimes_{(A \otimes A)} (\kappa(\mathfrak{m}) \otimes \kappa(\mathfrak{m})) = B/s(\mathfrak{m})B + t(\mathfrak{m})B$$

We conclude that

$$\mathcal{O}_{R'', e''} = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}} + t(\mathfrak{m})B_{\mathfrak{q}}$$

and so now we know that this ring has dimension d_1.

We claim this implies we can find an element $f \in \mathfrak{m}$ such that

$$\dim(B_{\mathfrak{q}}/(s(\mathfrak{m})B_{\mathfrak{q}} + fB_{\mathfrak{q}}) < d_2$$

Namely, suppose $\mathfrak{n}_j \supset s(\mathfrak{m})B_{\mathfrak{q}}$, $j = 1, \ldots, m$ correspond to the minimal primes of the local ring $B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$. There are finitely many as this ring is Noetherian (since it is essentially of finite type over a field – but also because a Cohen-Macaulay ring is Noetherian). By the Cohen-Macaulay condition we have $\dim(B_{\mathfrak{q}}/\mathfrak{n}_j) = d_2$, for

example by Algebra, Lemma 00NA. Note that $\dim(B_{\mathfrak{q}}/(\mathfrak{n}_j + t(\mathfrak{m})B_{\mathfrak{q}})) \leq d_1$ as it is a quotient of the ring $\mathcal{O}_{R'',e''} = B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}} + t(\mathfrak{m})B_{\mathfrak{q}}$ which has dimension d_1. As $d_1 < d_2$ this implies that $\mathfrak{m} \not\subset t^{-1}(\mathfrak{n}_i)$. By prime avoidance, see Algebra, Lemma 00DS, we can find $f \in \mathfrak{m}$ with $t(f) \notin \mathfrak{n}_j$ for $j = 1, \ldots, m$. For this choice of f we have the displayed inequality above, see Algebra, Lemma 00KW.

Set $A' = A/fA$ and $U' = \mathrm{Spec}(A')$. Then it is clear that $U' \to U$ is an immersion, locally of finite presentation and that $u \in U'$. Thus (1), (2) and (3) of the lemma hold. The morphism

$$U' \times_{g,U,t} R \longrightarrow U$$

factors through $\mathrm{Spec}(A)$ and corresponds to the ring map

$$B/t(f)B = A/(f) \otimes_{A,t} B \xleftarrow{s} A$$

Now, we see $t(f)$ is not a zerodivisor on $B_{\mathfrak{q}}/s(\mathfrak{m})B_{\mathfrak{q}}$ as this is a Cohen-Macaulay ring of positive dimension and f is not contained in any minimal prime, see for example Algebra, Lemma 02JN. Hence by Algebra, Lemma 046Z we conclude that $s : A_{\mathfrak{m}} \to B_{\mathfrak{q}}/t(f)B_{\mathfrak{q}}$ is flat with fibre ring $B_{\mathfrak{q}}/(s(\mathfrak{m})B_{\mathfrak{q}} + t(f)B_{\mathfrak{q}})$ which is Cohen-Macaulay by Algebra, Lemma 02JN again. This implies part (4) of the lemma. To see part (5) note that by Diagram (04MN) the fibre F'_u is equal to the fibre of h over u. Hence $\dim_{e'(u)}(F'_u) = \dim(B_{\mathfrak{q}}/(s(\mathfrak{m})B_{\mathfrak{q}} + t(f)B_{\mathfrak{q}}))$ by Morphisms, Lemma 02FX and the dimension of this ring is $d_2 - 1$ by Algebra, Lemma 02JN once more. This proves the final assertion of the lemma and we win. □

References